the tRNAs paly a very important role in protein synthesis. They act as molecular "bridges" that connect mRNA codons to the amino acids they encode. Each tRNA has an anticodon loop, which has a sequence of three nucleotides called an anticodon. They can bind to specific codon in the mRNA. The other end of the tRNA carries the amino acid specified by the codons.

6.4.3 Ribosomes

Ribosomes are the cellular organelles where polypeptides (proteins) are built. They themselves are made up of different proteins and rRNA. Each ribosome has two subunits, a large one and a small one, which sandwiches the mRNA as shown in figure 14

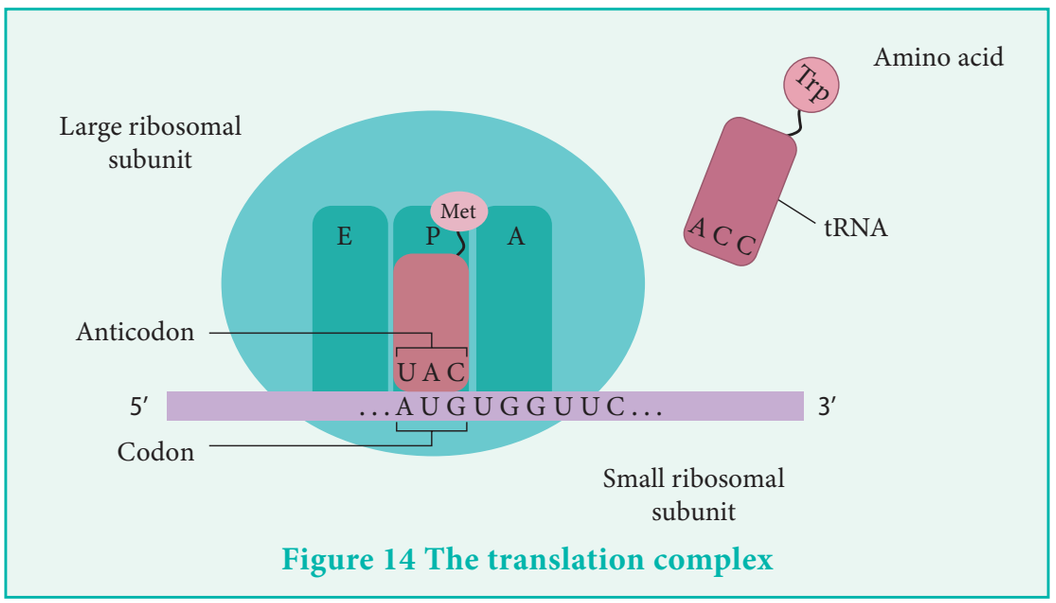

Figure 14 The translation complex

The ribosome provides three different structural packets where tRNA can enter and transfer its amnoacid to the growing polypeptide chain. These slots are called the A, P, and E sites. The ribosome itself can act as an enzyme, catalyzing the chemical reaction that links amino acids together to make a chain.

6.4.4 Molecular events in translation

The translation process can be divided into three different stages: initiation, elongation, and termination.

6.4.4.1 Translation Initiation

Inititaion require three important components

- A ribosome – the large subunit and the small subunit
- An mRNA with instructions for the encoding protein.
- An "initiator" tRNA carrying the first amino acid in the protein, which is almost always methionine (Met)

The methonine tRNA attaches to the small ribosomal subunit. Together, they bind to the 5' end of the mRNA by recognizing the 5' cap in eukaryotes and then they move along the mRNA in the 3' direction, stopping when they reach the start codon AUG. This is later

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